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Solutions

Base IMIS Field Data Collection User Manual - Coordinator

Integrated Municipal Information System (IMIS)

Innovative Solution Pvt. Ltd (ISPL)

Base IMIS Field Data Collection User Manual - Enumerator

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Table of Contents

Chapter 1	Introduction.....	1
Chapter 2	Mergin	2
2.1.	<i>Workflow</i>	<i>2</i>
2.2.	<i>Key Features of Mergin</i>	<i>3</i>
2.3.	<i>Signing in Mergin</i>	<i>3</i>
2.4.	<i>Home Page and Dashboard.....</i>	<i>3</i>
2.5.	<i>Sharing a project</i>	<i>5</i>
2.6.	<i>Project history and versions</i>	<i>6</i>
Chapter 3	QGIS	8
3.1.	<i>Introduction</i>	<i>8</i>
3.2.	<i>Connection with Mergin account</i>	<i>8</i>
3.3.	<i>Downloading an existing project.....</i>	<i>9</i>
3.4.	<i>Syncing with Mergin account</i>	<i>10</i>
3.5.	<i>Viewing data using Attribute Table</i>	<i>10</i>
3.6.	<i>Downloading Data.....</i>	<i>11</i>

Chapter 1 INTRODUCTION

This is an operational training manual for survey coordinators. This manual will give a brief idea about what the survey coordinator should do, how to use the Mergin Web app as well as QGIS to monitor the process of data collection.

The key responsibility of the survey coordinator is to mobilize surveyors in the field for data collection and to coordinate with field supervisors to ensure the timely completion of survey, ensure optimal quality field data collection with regular monitoring and make necessary backups of field data during the survey. Furthermore, they also have to communicate with local authorities to gain their cooperation during the survey and manage all necessary supplies/logistics required during the field survey. They will be provided with full access of the survey project through the Mergin web app. Another key responsibility is to provide access to enumerators through the Mergin web app and assign proper roles as well. The key tasks of the survey coordinator are:

- a. Provide access to enumerator and supervisors of the developed project through Mergin.
- b. Setup QGIS in the coordinator's workstation.
- c. Periodically monitor, synchronize, and download data collected from QGIS.

The training will be conducted by the development team. A person must have a bachelor's degree in statistics, computer science or similar academic background, and have experience of conducting and managing survey data to be eligible to work as a Survey coordinator. Furthermore, they must have good communication skills as well knowledge of using mobile devices/ using the internet. Survey coordinators must be available for communication through mobile throughout the day, must have mobile internet access as well as internet access at home for monitoring survey status and quality control.

Chapter 2 MERGIN

Mergin is a repository of geo-data for collaborative work. It is a service to synchronize various data between different users and platforms, so changes made by other users are updated within the files and folders. Mergin allows the owner of the project to give access permissions to read or write. The changes that were made by the different users can be traced and consolidated in Mergin which allows the owner to view the changes and also retrieve previous versions of data.

2.1. Workflow

After registering with the Mergin service, you will need to set up a QGIS project on your desktop. You can then upload your project and associated data to Mergin, either through the website, or via Mergin plugin in QGIS. The project can be accessed on Input mobile application. You can start collecting data and upload back the changes to the Mergin services.



Figure 1-1 Figure 1. Workflow of Mergin

2.2. Key Features of Mergin

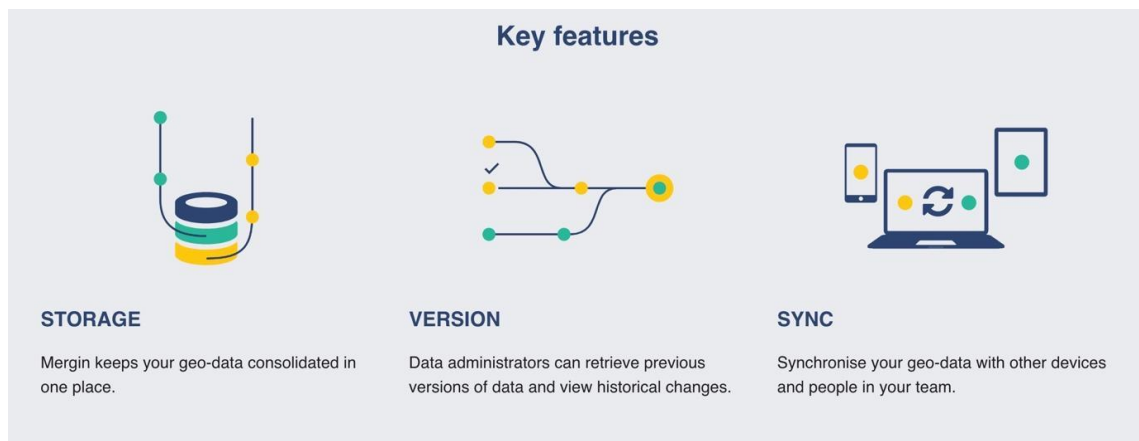
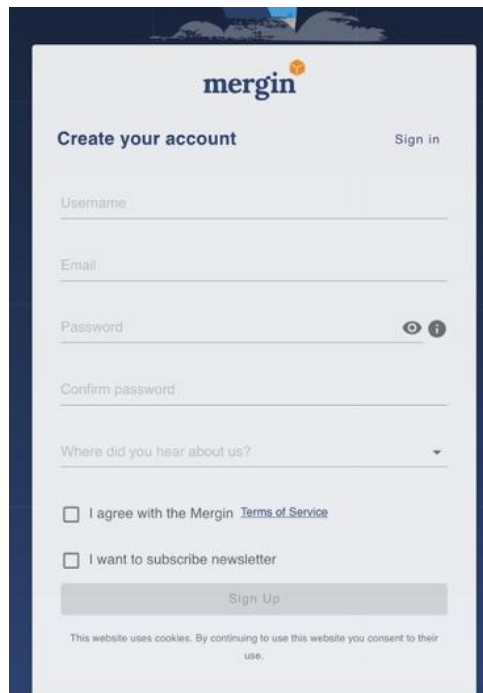


Figure 1-2 Figure 2. Features of Mergin (Source: Mergin Website)

2.3. Signing in Mergin

Signing up in Mergin is free, the user needs to enter their email address and then enter username and password. After that an activation link will be sent to the users' email which when clicked confirms the authenticity of the email address. The user is then good to go, then can log into their Mergin account.



The screenshot shows the 'Create your account' form on the Mergin website. The form includes fields for Username, Email, Password, and Confirm password. There is a 'Sign in' link at the top right. Below the password fields, there is a dropdown menu for 'Where did you hear about us?'. At the bottom, there are two checkboxes: 'I agree with the Mergin Terms of Service' and 'I want to subscribe newsletter'. A 'Sign Up' button is located below the checkboxes. At the very bottom, there is a small disclaimer: 'This website uses cookies. By continuing to use this website you consent to their use.'

Figure 1-3 Figure 3. Sign in Mergin

2.4. Home Page and Dashboard

On the left panel of the webpage, the user will be able to access profiles, organizations, and subscription plans.

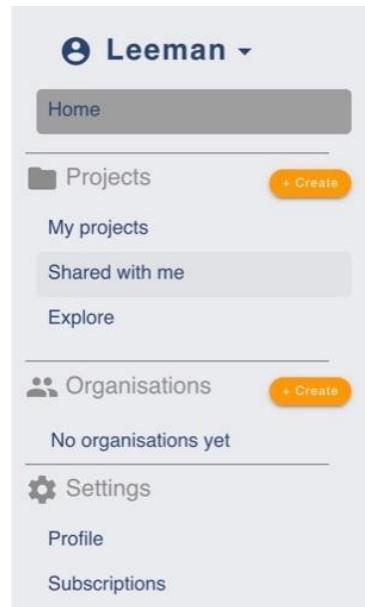


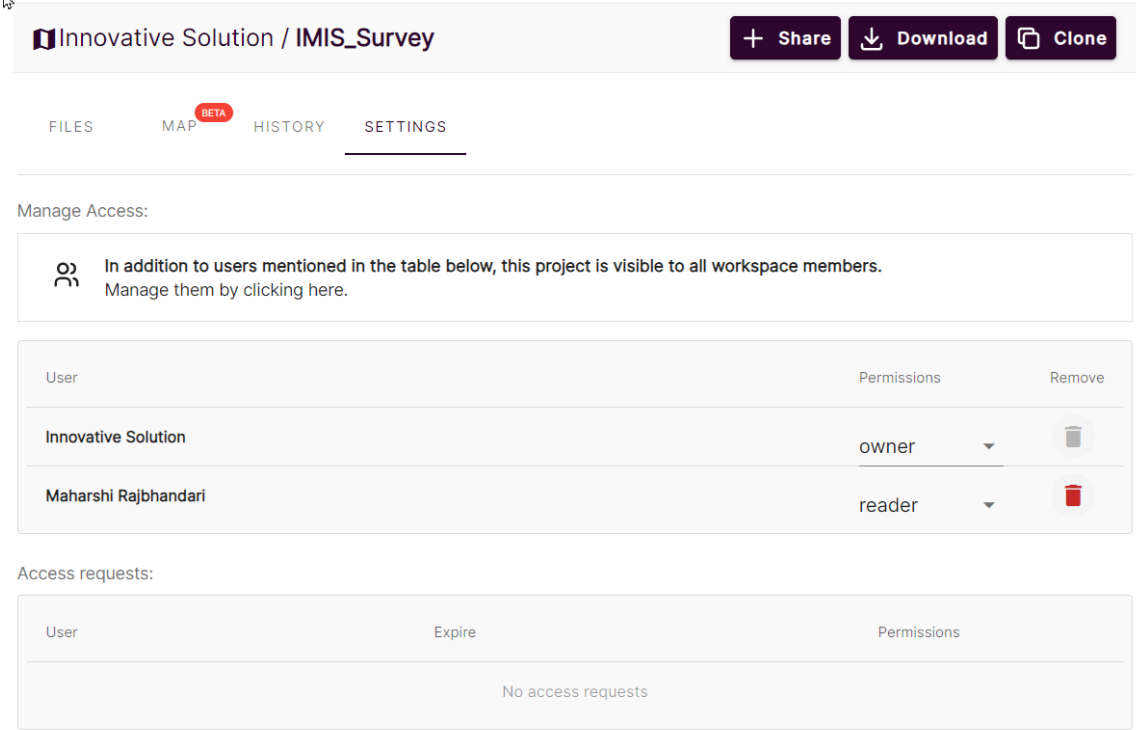
Figure 1-4 Figure 4. Left panel

- a. Under Projects
 - i. To create a new project, select +Create in front of Projects
 - ii. My projects list all projects the users have created
 - iii. Shared with me lists the projects of others shared with the user
 - iv. Explore lists all the public projects.
- b. Under Organizations
 - i. To create a new organization, select +Create in front of Organization
 - ii. Click on the name to change the organization.
- c. Under Settings
 - i. Profile displays the account details of the users, email notification settings and invitations to join organizations.
 - ii. Subscriptions: displays user plans, payment history and invoices.

2.5. Sharing a project

Mergin allows the users to share their project with other users for them to contribute to the project. The coordinator is responsible for providing access to the project to the different enumerators that will be carrying out the survey. The coordinator should collect the Mergin username, add them to the project and assign them the proper permissions through the dropdown. The permission that is generally assigned to the enumerator is the Write permission which allows the user to write on the project. The project can be shared to other enumerators by:

- i. Select the desired project. Then on the Settings tab of that project.



The screenshot shows the 'Settings' tab for a project named 'IMIS_Survey'. At the top, there are buttons for '+ Share', 'Download', and 'Clone'. Below these are tabs for 'FILES', 'MAP', 'HISTORY', and 'SETTINGS' (which is active). The 'Manage Access' section shows a message: 'In addition to users mentioned in the table below, this project is visible to all workspace members. Manage them by clicking here.' Below this is a table with columns 'User', 'Permissions', and 'Remove'.

User	Permissions	Remove
Innovative Solution	owner	
Maharshi Rajbhandari	reader	

Below the table is the 'Access requests' section, which currently shows 'No access requests'.

- ii. Select the Share Button and enter the registered email address of the enumerator.

The three permissions are:

- a. Owner (read/write/delete/share)
- b. Writer (read/write/delete)
- c. Reader (read)

The enumerators need to be provided with Writer access for data collection. Additionally, a user can be provided Reader access to view the data for monitoring and supervision of the collected data.

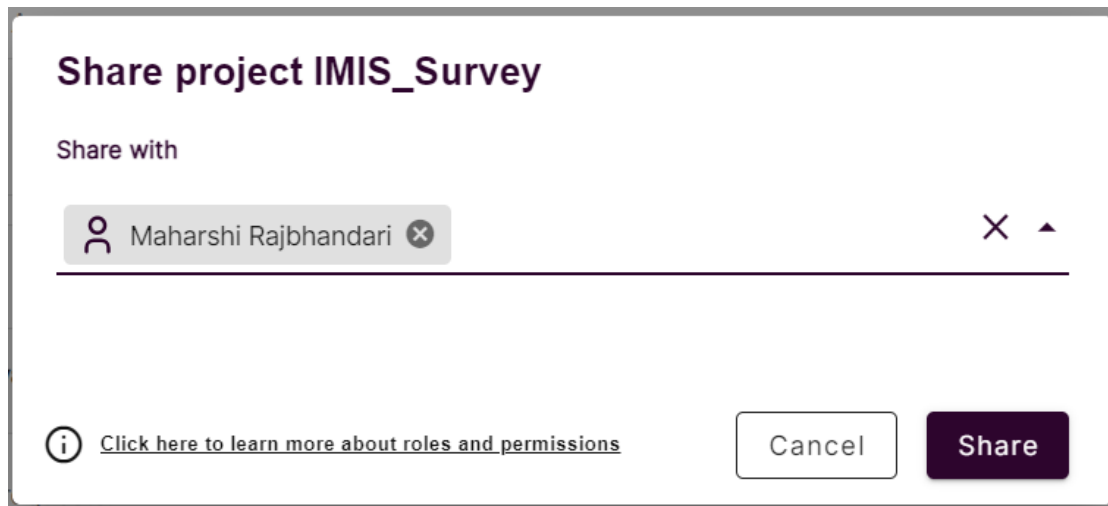


Figure 1-5 Figure 5. Sharing project

2.6. Project history and versions

Mergin allows the users to see the details of the changes made to a project. Every detail from the devices used, to the users who change anything can be seen. Each time a user syncs the project from mobile device or from QGIS, a new version is made. This will help the coordinator to look for earlier versions and revert back to the old version due to some technical issues in the application in the newer version, if needed.

To view the project history:

- i. Click on the History tab in the project page.
- ii. New tab will open with:
 - a. Version
 - b. Created
 - c. Author
 - d. Number of files added, deleted or updated
 - e. Size
 - f. Download

FILES HISTORY SETTINGS

Version	Created	Author				Size	
v89	2 days ago	Leeman	0	0	1	16.82 MB	
v88	2 days ago	Leeman	0	0	2	16.82 MB	
v87	1 week ago	Leeman	1	0	1	16.83 MB	
v86	1 week ago	Sukuchha	0	0	1	15.79 MB	
v85	1 week ago	Sukuchha	0	0	5	15.79 MB	
v84	1 week ago	Leeman	0	0	1	15.78 MB	
v83	1 week ago	Leeman	1	0	2	15.79 MB	
v82	1 week ago	Sukuchha	0	0	1	15.77 MB	

Figure 1-6 Figure 6. View History

Chapter 3 QGIS

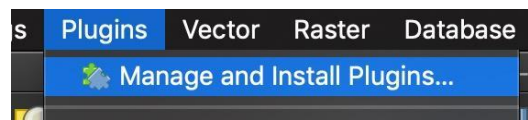
3.1. Introduction

QGIS is a free and open-source cross-platform desktop geographic information system (GIS) application that supports viewing, editing, and analysis of geospatial data.

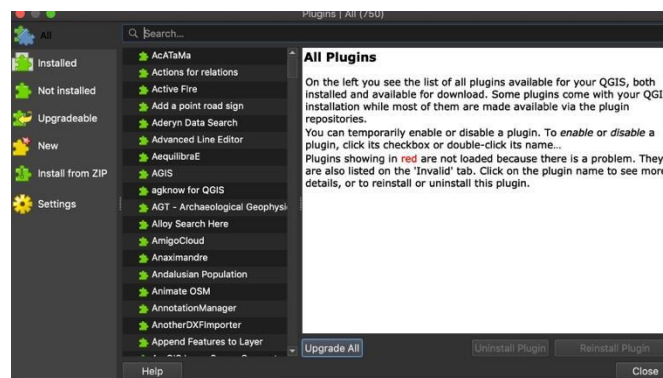
3.2. Connection with Mergin account

To install Mergin plugin in QGIS:

- i. Open QGIS
- ii. From the main menu, go to 'Plugins' and go to 'Manage and Install plugins'



- iii. A new window will appear



- iv. In the search section, type: Mergin
- v. Select Mergin in the filtered section and then click 'Install Plugin'
- vi. Once installed, you should see Mergin in the QGIS Browser panel.

Note: If you cannot see your Browser panel, you can enable it by going to the QGIS main menu> View> Panels> Browser

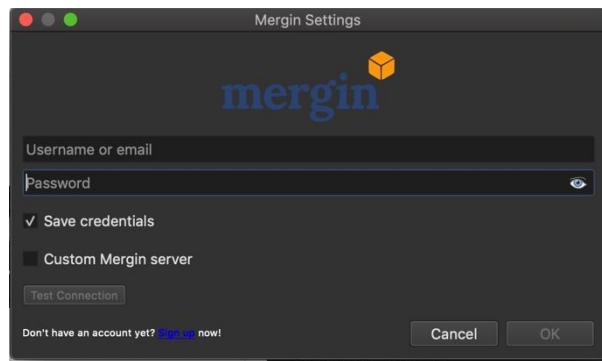
- vii. A new toolbar should also appear in your QGIS toolbar.



After installation, the user needs to configure the plugin by entering the Mergin username and password.

- viii. From the Mergin toolbar, click on configure

- ix. A new window will appear, type the username and password



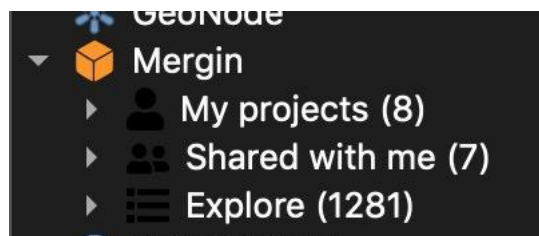
The Mergin is then connected with QGIS.

3.3. Downloading an existing project

The consultant will provide a completed project to the coordinator that will be shared via Mergin. The shared project can be found under the Shared with Me section, through which the coordinator can access and download the project to their workstation.

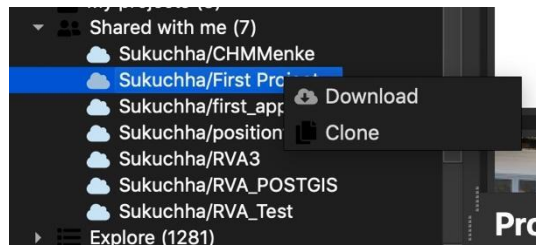
There are different sections under the Mergin in the Browser Panel:

- i. My Project
- ii. Shared with me
- iii. Explore



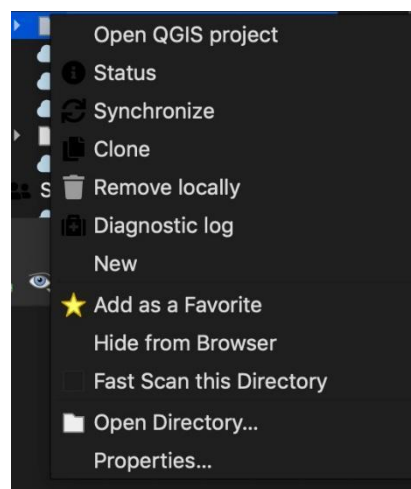
Multiple projects will be listed under various headings. In order to access the project that has been shared with the coordinator, the project is to be located under the Shared With me tab. To download a project:

- i. Right click on the project and select Download.
- ii. A new window will appear to save the project locally on the PC. Browse to the folder you want to download the project and click Select folder.
- iii. Once the download is complete, you will be presented with an option to open the project in QGIS. Selecting Yes will open the project and all the associated layers.



3.4. Syncing with Mergin account

Synchronization is the process of merging and matching data from different devices so that all projects are on the same page. This is to be done periodically by the coordinator to merge and update the changes to their local workstation. This is a crucial task as data collected by different enumerators cannot be viewed by the coordinator unless the changes have been synced. Synchronizing a project consist of two major parts:



i. Status

It is recommended to run the project status after changing your layers and project. To see the status of the project and data, from the Mergin plugin, click on *Status*. This will show all the status such as things that were added, updated, and deleted into the layers.

ii. Synchronization

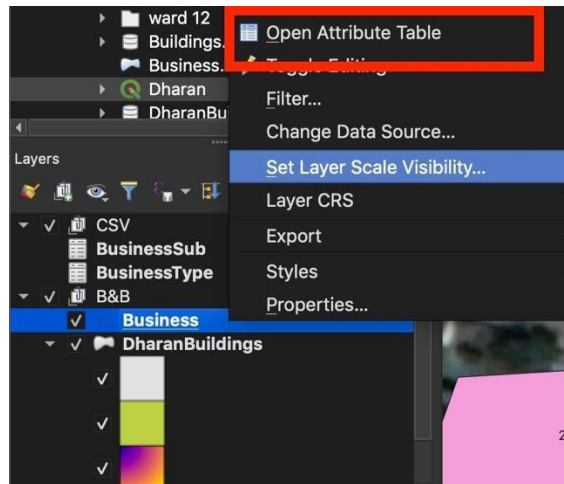
Once the user is satisfied with the status of the project, synchronization can be done. Click on the *Synchronize* icon from the Mergin toolbar to sync the project and data. You can also check the status and sync the project by Right-clicking the project.

Note: Synchronizing data works both ways, all the changes will be uploaded to the server and pending changes will be downloaded locally.

3.5. Viewing data using Attribute Table

The attribute table displays the attribute/information on features of a selected layer. The coordinator can view the data that has been collected through this and periodically synchronize the data and examine the data collected from the field periodically. Additionally, the coordinator must also inform or organize a reorientation of the questionnaire (if required) to the enumerators if any issues are found in the data that has been collected. To view the attribute table:

- i. On the Layers menu, right click a layer.
- ii. Choose the Open Attribute Table option.



3.6. Downloading Data

The coordinator can download the CSV file containing the collected data in the attribute table, if required. To download data from Attribute table:

- i. Right-click the attribute layer in the layer's menu
- ii. Select Export and Save features as...
- iii. At the top of the program window, select Comma Separated Value [CSV] in the format drop-down menu.
- iv. Type a file name for the file to be saved.
- v. Click the three-dot button next to the file name to choose a path to where the file should be saved.
- vi. Uncheck any fields you do not want in your resulting file.
- vii. Optionally (probably) uncheck the add saved file to map at the bottom of the window so that the file does not get added to your project as a data layer.
- viii. Click the green check mark, OK button.

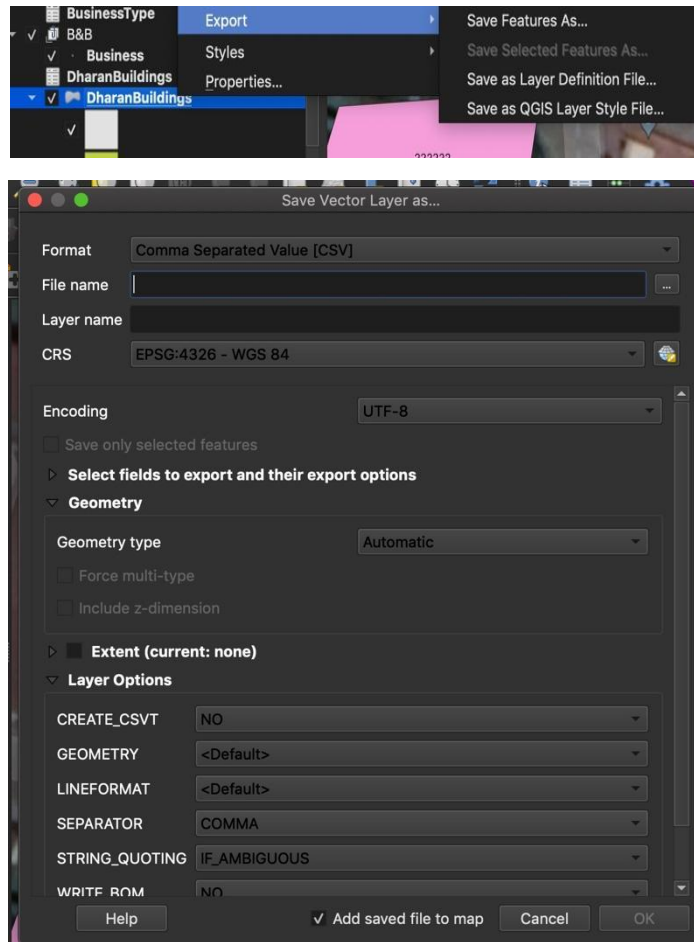


Figure Downloading Data